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Title :

Feedforward and Adaptive Correction of Time-Varying Thermal Bias for Coarse Acquisition in Optical Communication Systems

Co-author :

Kazuki Takashima, Yuki Kusano, Satoshi Ikari, Ryu Funase

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Abstract :

In optical communication systems, the performance of pointing, acquisition, and tracking (PAT) is critical for reliable link establishment. During the coarse acquisition phase, initial pointing error is a dominant factor determining acquisition success probability and scan time. While residual errors can be compensated through feedback in the fine tracking phase, systematic pointing biases remain uncorrected at the beginning of acquisition. This issue is particularly critical in LEO systems, where narrow beam divergence and rapidly varying thermal environments associated with compact satellite platforms make initial pointing errors a critical contributor to acquisition performance.

This study addresses time-varying line-of-sight (LOS) bias errors induced by thermal distortion. In LEO satellites, periodic and irregular temperature variations caused by repeated eclipse cycles lead to structural deformation, resulting in LOS biases that vary with orbital phase. Thermo-structural analysis using Thermal Desktop and Femap indicates that deformations on the order of 100–200 μm can occur in meter-scale satellite structures, corresponding to LOS biases on the order of tens to hundreds of micro radians. This magnitude is comparable to the beam divergence and LOS errors caused by attitude determination accuracy, and thus has a non-negligible impact on acquisition performance.

Conventional approaches primarily rely on observation-based corrections, while the integration of physics-based thermal modeling with on-orbit adaptive correction has not been sufficiently explored.

To address this issue, this study proposes a two-layer correction framework that integrates physics-based prediction with adaptive correction using on-orbit observations. First, a feedforward model is constructed based on thermo-structural analysis using Thermal Desktop and Femap to predict LOS bias as a function of orbital conditions and operational states, and is applied to correct the initial pointing direction at the start of coarse acquisition. Second, pointing error and received optical power information obtained from PAT sensors, such as quadrant detectors or focal plane modules, are used to iteratively adjust the correction based on the discrepancy between prediction and observation, thereby compensating for model errors and temporal variations.

The proposed method is evaluated through high-fidelity numerical simulations incorporating representative LEO thermal environments. Performance is assessed in terms of acquisition success rate, acquisition time, and required scan area, and compared with conventional approaches such as no correction, static bias compensation, and feedback-only methods based solely on observational data, thereby explicitly evaluating the benefit of incorporating a priori thermal models.

The novelty of this work lies in explicitly treating thermally induced pointing bias as a predictable, time-varying component and directly linking it to acquisition performance improvement. By reducing initial pointing uncertainty and required scan area in coarse acquisition, the proposed framework improves acquisition efficiency and robustness, particularly in LEO optical communication systems for small and micro satellites.